

Statistical Modeling in R: Multiple Linear Regression and ANCOVA

Two case studies in regression modeling, model selection, and diagnostics

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Project Overview

This project presents two statistical modeling case studies in R. The first uses multiple linear regression to explain log prostate-specific antigen (`lpsa`) from clinical covariates. The second uses ANCOVA to compare infusion systems while modeling alarm delay as a function of inverse flow rate.

The analysis covers exploratory data analysis, model specification, manual estimation, inference, variable selection, nested-model comparison, and model diagnostics.

Case Study 1 — Multiple Linear Regression on Prostate Cancer Data

Exploratory Data Analysis

```
Prostate = read.table("Prostate.txt", sep=";", header=T, dec=".")
head(Prostate)
```

```
##      lccavol  lweight age      lbph svi      lcp gleason pgg45      lpsa
## 1 -0.5798185 2.769459  50 -1.386294  0 -1.386294      6      0 -0.4307829
## 2 -0.9942523 3.319626  58 -1.386294  0 -1.386294      6      0 -0.1625189
## 3 -0.5108256 2.691243  74 -1.386294  0 -1.386294      7     20 -0.1625189
## 4 -1.2039728 3.282789  58 -1.386294  0 -1.386294      6      0 -0.1625189
## 5  0.7514161 3.432373  62 -1.386294  0 -1.386294      6      0  0.3715636
## 6 -1.0498221 3.228826  50 -1.386294  0 -1.386294      6      0  0.7654678
```

```
dim(Prostate)
```

```
## [1] 97  9
```

The `Prostate` data frame contains 97 observations and 9 variables.

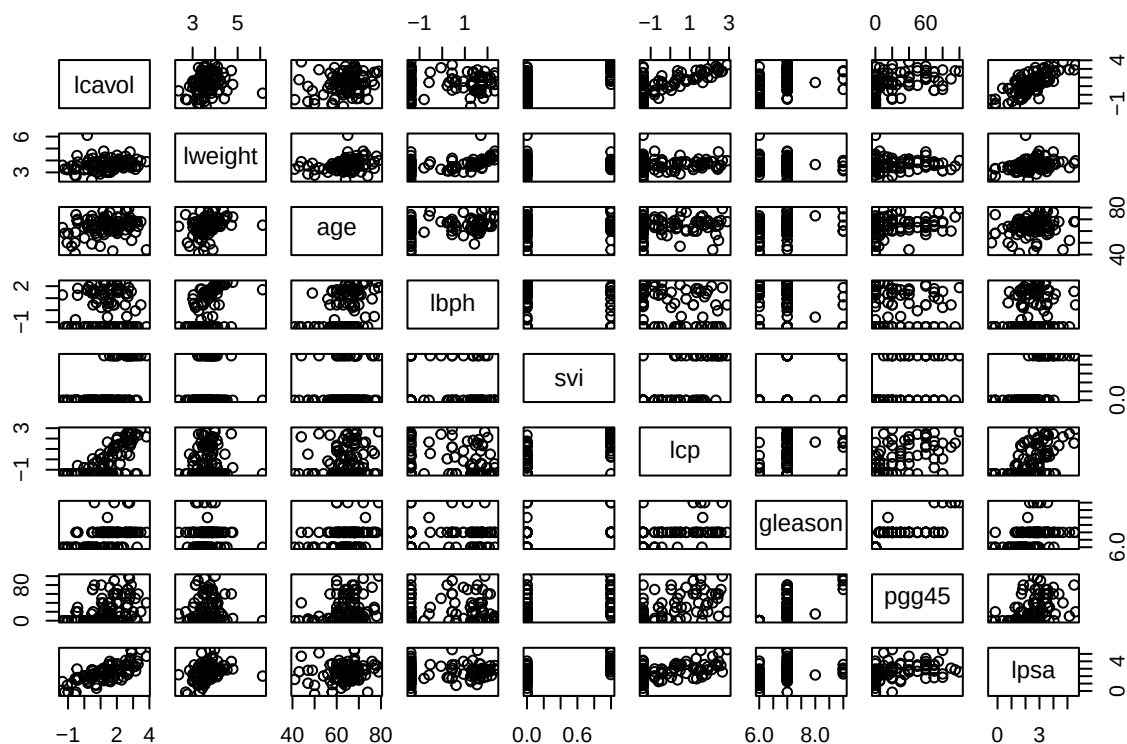
```
str(Prostate)
```

```
## 'data.frame': 97 obs. of 9 variables:
## $ lcavol : num -0.58 -0.994 -0.511 -1.204 0.751 ...
## $ lweight: num 2.77 3.32 2.69 3.28 3.43 ...
## $ age : int 50 58 74 58 62 50 64 58 47 63 ...
## $ lbph : num -1.39 -1.39 -1.39 -1.39 -1.39 ...
## $ svi : int 0 0 0 0 0 0 0 0 0 0 ...
## $ lcp : num -1.39 -1.39 -1.39 -1.39 -1.39 ...
## $ gleason: int 6 6 7 6 6 6 6 6 6 6 ...
## $ pgg45 : int 0 0 20 0 0 0 0 0 0 0 ...
## $ lpsa : num -0.431 -0.163 -0.163 -0.163 0.372 ...
```

```
summary(Prostate)
```

```
##      lcavol      lweight      age      lbph
## Min.   :-1.3471  Min.   :2.375  Min.   :41.00  Min.   : -1.3863
## 1st Qu.: 0.5128  1st Qu.:3.376  1st Qu.:60.00  1st Qu.: -1.3863
## Median : 1.4469  Median :3.623  Median :65.00  Median : 0.3001
## Mean   : 1.3500  Mean   :3.653  Mean   :63.87  Mean   : 0.1004
## 3rd Qu.: 2.1270  3rd Qu.:3.878  3rd Qu.:68.00  3rd Qu.: 1.5581
## Max.   : 3.8210  Max.   :6.108  Max.   :79.00  Max.   : 2.3263
##      svi      lcp      gleason      pgg45
## Min.   :0.0000  Min.   : -1.3863  Min.   :6.000  Min.   : 0.00
## 1st Qu.:0.0000  1st Qu.: -1.3863  1st Qu.:6.000  1st Qu.: 0.00
## Median :0.0000  Median : -0.7985  Median :7.000  Median :15.00
## Mean   :0.2165  Mean   : -0.1794  Mean   :6.753  Mean   :24.38
## 3rd Qu.:0.0000  3rd Qu.: 1.1787  3rd Qu.:7.000  3rd Qu.:40.00
## Max.   :1.0000  Max.   : 2.9042  Max.   :9.000  Max.   :100.00
##      lpsa
## Min.   : -0.4308
## 1st Qu.: 1.7317
## Median : 2.5915
## Mean   : 2.4784
## 3rd Qu.: 3.0564
## Max.   : 5.5829
```

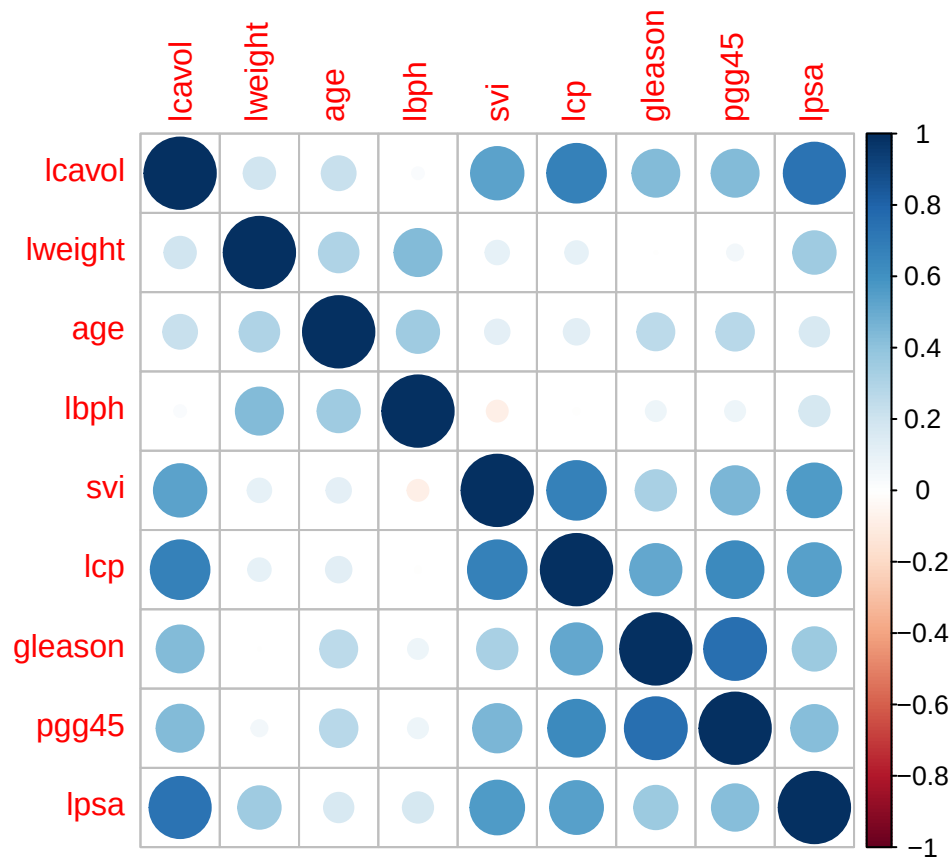
```
pairs(Prostate)
```



```
cor(Prostate)
```

```
##          lcavol      lweight      age      lbph      svi      lcp
## lcavol  1.0000000  0.194128286  0.2249999  0.027349703  0.53884500  0.675310484
## lweight 0.1941283  1.000000000  0.3075286  0.434934636  0.10877851  0.100237795
## age     0.2249999  0.307528614  1.0000000  0.350185896  0.11765804  0.127667752
## lbph    0.0273497  0.434934636  0.3501859  1.000000000 -0.08584324 -0.006999431
## svi     0.5388450  0.108778505  0.1176580 -0.085843238  1.00000000  0.673111185
## lcp     0.6753105  0.100237795  0.1276678 -0.006999431  0.67311118  1.000000000
## gleason 0.4324171 -0.001275658  0.2688916  0.077820447  0.32041222  0.514830063
## pgg45   0.4336522  0.050846821  0.2761124  0.078460018  0.45764762  0.631528245
## lpsa    0.7344603  0.354120390  0.1695928  0.179809410  0.56621822  0.548813169
##          gleason      pgg45      lpsa
## lcavol  0.432417056  0.43365225  0.7344603
## lweight -0.001275658  0.05084682  0.3541204
## age     0.268891599  0.27611245  0.1695928
## lbph    0.077820447  0.07846002  0.1798094
## svi     0.320412221  0.45764762  0.5662182
## lcp     0.514830063  0.63152825  0.5488132
## gleason 1.000000000  0.75190451  0.3689868
## pgg45   0.751904512  1.00000000  0.4223159
## lpsa    0.368986803  0.42231586  1.0000000
```

```
library(corrplot)
corrplot(cor(Prostate))
```



Univariate Analysis The response variable is `lpsa`. Most predictors are quantitative, while `svi` is binary and `gleason` takes only a small number of discrete values.

The `lpsa` variable ranges from -0.43 to 5.58, with a mean of approximately 2.48. The variables `lccavol`, `lweight`, `lbph`, and `lcp` are logarithmic transformations. The variable `svi` equals 1 for only 21 of the 97 observations, meaning that most patients do not present seminal vesicle invasion. The variable `pgg45` is highly dispersed, with values ranging from 0 to 100.

Bivariate Analysis The correlation analysis shows that `lccavol` is the variable most strongly correlated with `lpsa`, with a correlation of approximately 0.73. This indicates a strong positive association between tumor volume and PSA level. The variables `svi`, `lcp`, `pgg45`, `gleason`, and `lweight` are also positively correlated with `lpsa`, although more moderately.

By contrast, `age` and `lbph` are only weakly correlated with `lpsa`, with correlations of approximately 0.17 and 0.18. They therefore appear to have a weaker direct relationship with the response variable.

There are also substantial correlations among some predictors, particularly between `gleason` and `pgg45`, between `lccavol` and `lcp`, and between `svi` and `lcp`. This suggests potential redundancy among predictors in a multiple regression model. Predictor selection should therefore not rely only on marginal correlations with `lpsa`; correlations among the predictors themselves must also be considered.

The scatterplot matrix confirms that `lpsa` is most clearly positively associated with `lccavol`. The scatterplot of `lpsa` against `lccavol` displays a clear increasing trend: as `lccavol` increases, `lpsa` tends to increase as well.

Positive trends are also visible between `lpsa` and several other variables, including `svi`, `lcp`, `pgg45`, `gleason`, and `lweight`, although these relationships are less pronounced than the relationship with `lcavol`.

Because `svi` is binary, plots involving `svi` form two vertical or horizontal bands corresponding to values 0 and 1. This reflects the fact that `svi` is not a continuous quantitative variable.

The variable `gleason` also takes only a small number of values, mainly 6 and 7, so the corresponding scatterplots appear in bands as well.

Overall, `pairs(Prostate)` visually confirms the main conclusions from the correlation matrix: `lcavol` is the predictor most strongly associated with `lpsa`, while several predictors are correlated with one another. This dependence must be taken into account when selecting a regression model.

Model Specification

The objective is to explain `lpsa` using the remaining variables. We consider the following multiple linear regression model:

$$lpsa_i = \beta_0 + \beta_1 lcavol_i + \beta_2 lweight_i + \beta_3 age_i + \beta_4 lbph_i + \beta_5 svi_i + \beta_6 lcp_i + \beta_7 gleason_i + \beta_8 pgg45_i + \varepsilon_i$$

The term ε_i represents the random error associated with observation i . We assume that the errors are independent, centered, and have constant variance.

Manual OLS Estimation

```
X=as.matrix(Prostate[,1:8])
X=cbind(intercept=1,X)
Y=as.matrix(Prostate$lpsa,ncol=1)
beta.est=solve(t(X)%*%X)%*% t(X)%*%Y
n=length(Y)
p=ncol(X)
sigma.est=sqrt(sum( (X%*%beta.est -Y)^2 )/(n-p))
V= solve(t(X)%*%X) *sigma.est^2
stddev=sqrt(diag(V))
```

In the code above, `beta.est` and `sigma.est` are the estimators used for the linear regression model.

Using all eight predictors, the estimated full model is:

$$\widehat{lpsa} = 0.6694 + 0.5870 lcavol + 0.4545 lweight - 0.0196 age + 0.1071 lbph + 0.7662 svi - 0.1055 lcp + 0.0451 gleason + 0.0045 pgg45$$

```
t.stat = beta.est/stddev
p.val = 2 * pt(-abs(t.stat), df = n - p)
resultats.tests = data.frame(
  beta_estime = as.vector(beta.est),
  ecart_type = stddev,
  statistique_t = t.stat,
  p_valeur = p.val,
  significatif_5 = ifelse(p.val < 0.05, "oui", "non")
)
resultats.tests
```

	beta_estime	ecart_type	statistique_t	p_valeur	significatif_5
## intercept	0.669399027	1.296381277	0.5163597	6.068984e-01	non
## lcavol	0.587022881	0.087920374	6.6767560	2.110634e-09	oui
## lweight	0.454460641	0.170012071	2.6731081	8.956206e-03	oui
## age	-0.019637208	0.011172743	-1.7575995	8.229321e-02	non
## lbph	0.107054351	0.058449332	1.8315753	7.039819e-02	non
## svi	0.766155885	0.244309492	3.1360054	2.328823e-03	oui
## lcp	-0.105473570	0.091013484	-1.1588785	2.496408e-01	non
## gleason	0.045135964	0.157464467	0.2866422	7.750601e-01	non
## pgg45	0.004525324	0.004421185	1.0235545	3.088513e-01	non

From the test statistics and p-values for the model coefficients, only three predictors appear significant at the 5% level: `lcavol`, `lweight`, and `svi`.

We then compute the estimated log prostate-specific antigen level for the covariate values of the second observation:

```
Yhat2 = X[2, ] %*% beta.est
Yhat2
```

```
##           [,1]
## [1,] 0.7240557
```

For all observations, the fitted values are:

```
Yhat = X %*% beta.est
Yhat
```

```
##           [,1]
## 1  0.8744063
## 2  0.7240557
## 3  0.5437102
## 4  0.5842042
## 5  1.7214934
## 6  0.8072678
## 7  1.9068186
## 8  2.1274633
## 9  1.1678020
## 10 1.3063870
## 11 1.4490247
## 12 0.8296163
## 13 2.1195052
## 14 1.9197316
## 15 2.0578787
## 16 1.9377909
## 17 1.2967266
## 18 2.4584438
## 19 1.2879468
## 20 1.7348576
## 21 2.0069463
## 22 2.7132249
## 23 1.0567903
## 24 2.5568346
```

25 1.7955333
26 2.0525960
27 1.9529918
28 1.6226867
29 2.0830775
30 2.2933055
31 1.9843831
32 2.8751959
33 1.9568477
34 1.3684808
35 1.0949261
36 2.7701955
37 1.9252982
38 1.1420780
39 3.9469131
40 1.8631297
41 2.0494303
42 2.2132893
43 2.0503229
44 2.3916288
45 2.4343770
46 2.5861067
47 4.0810432
48 2.5765411
49 2.7081115
50 2.1351206
51 2.3300147
52 2.9506919
53 2.1523994
54 2.9891960
55 3.2963100
56 2.8194686
57 1.7534190
58 2.2596815
59 2.1115833
60 2.6265233
61 2.2266838
62 3.5302967
63 3.0102254
64 3.6939326
65 2.5447554
66 2.6645755
67 2.9683847
68 3.0406514
69 1.3245863
70 2.8007164
71 3.2841975
72 2.1192500
73 2.7667653
74 3.5438975
75 3.6471019
76 3.7088731
77 3.2127938
78 3.3629007

```
## 79 3.4749092
## 80 3.1559591
## 81 2.1637224
## 82 2.9931510
## 83 3.5608718
## 84 3.3670615
## 85 2.6420090
## 86 3.9104558
## 87 3.0070941
## 88 3.0373053
## 89 4.1306183
## 90 3.2095744
## 91 3.3725719
## 92 4.0702321
## 93 3.8229781
## 94 4.7053043
## 95 3.6175153
## 96 4.1878897
## 97 4.0918913
```

As expected, $\hat{Y}[2] = \hat{Y}2$.

Estimation with `lm()`

```
model8=lm(lpsa~.,data=Prostate)
summary(model8)
```

```
##
## Call:
## lm(formula = lpsa ~ ., data = Prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.73316 -0.37133 -0.01702  0.41414  1.63811
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.669399   1.296381   0.516  0.60690
## lcavol       0.587023   0.087920   6.677 2.11e-09 ***
## lweight      0.454461   0.170012   2.673  0.00896 **
## age         -0.019637   0.011173  -1.758  0.08229 .
## lbph         0.107054   0.058449   1.832  0.07040 .
## svi          0.766156   0.244309   3.136  0.00233 **
## lcp         -0.105474   0.091013  -1.159  0.24964
## gleason      0.045136   0.157464   0.287  0.77506
## pgg45        0.004525   0.004421   1.024  0.30885
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7084 on 88 degrees of freedom
## Multiple R-squared:  0.6548, Adjusted R-squared:  0.6234
## F-statistic: 20.86 on 8 and 88 DF,  p-value: < 2.2e-16
```

```
names(model8)
```

```
## [1] "coefficients" "residuals"      "effects"      "rank"
## [5] "fitted.values" "assign"         "qr"           "df.residual"
## [9] "xlevels"      "call"          "terms"        "model"
```

```
model8$coef
```

```
## (Intercept)      lcavol      lweight      age      lbph      svi
## 0.669399027 0.587022881 0.454460641 -0.019637208 0.107054351 0.766155885
##          lcp      gleason      pgg45
## -0.105473570 0.045135964 0.004525324
```

To verify the manual calculations, we compare the coefficients, residuals, and fitted values obtained from the matrix formulas with those returned by R's `lm()` function.

Coefficients:

```
model8$coef-beta.est
```

```
##          [,1]
## intercept -4.707346e-14
## lcavol     -1.909584e-14
## lweight    1.088019e-14
## age        1.984524e-15
## lbph       -5.093148e-15
## svi        3.108624e-15
## lcp        8.756884e-15
## gleason    1.334349e-14
## pgg45      -2.879641e-16
```

Residuals:

```
Y-Yhat-model8$residuals
```

```
##          [,1]
## 1  1.787459e-13
## 2  1.941780e-13
## 3  2.192690e-13
## 4  2.028377e-13
## 5  1.745271e-13
## 6  1.821043e-13
## 7  1.694200e-13
## 8  1.560974e-13
## 9  1.755956e-13
## 10 1.851297e-13
## 11 1.923184e-13
## 12 2.049472e-13
## 13 1.675327e-13
## 14 1.780798e-13
## 15 1.774136e-13
```

16 1.639799e-13
17 2.118028e-13
18 1.720846e-13
19 1.569855e-13
20 1.910694e-13
21 1.617595e-13
22 1.638689e-13
23 1.900702e-13
24 1.668665e-13
25 1.834227e-13
26 1.623146e-13
27 1.866285e-13
28 2.026990e-13
29 1.710021e-13
30 1.761369e-13
31 1.853726e-13
32 2.066125e-13
33 1.653261e-13
34 1.715295e-13
35 1.944001e-13
36 1.700862e-13
37 2.311207e-13
38 1.814104e-13
39 1.791900e-13
40 1.681988e-13
41 1.865452e-13
42 1.868922e-13
43 1.754152e-13
44 1.898898e-13
45 1.693506e-13
46 1.541128e-13
47 2.120526e-13
48 1.806480e-13
49 1.192518e-13
50 1.750822e-13
51 1.859068e-13
52 1.537659e-13
53 1.766365e-13
54 1.801337e-13
55 1.371125e-13
56 1.739026e-13
57 1.636469e-13
58 1.444400e-13
59 2.016165e-13
60 1.596223e-13
61 1.961764e-13
62 1.759703e-13
63 1.957878e-13
64 1.764144e-13
65 1.558753e-13
66 1.666167e-13
67 1.690037e-13
68 1.750960e-13
69 2.227107e-13

```
## 70 2.003120e-13
## 71 1.746381e-13
## 72 1.861844e-13
## 73 2.098322e-13
## 74 1.775247e-13
## 75 1.910139e-13
## 76 1.694200e-13
## 77 1.773304e-13
## 78 1.752626e-13
## 79 1.865730e-13
## 80 1.577072e-13
## 81 1.711964e-13
## 82 1.522116e-13
## 83 1.954695e-13
## 84 1.836586e-13
## 85 1.670886e-13
## 86 1.632028e-13
## 87 1.310063e-13
## 88 1.784128e-13
## 89 1.900147e-13
## 90 1.921796e-13
## 91 1.471046e-13
## 92 1.402628e-13
## 93 1.749711e-13
## 94 1.225964e-13
## 95 1.614264e-13
## 96 1.558753e-13
## 97 1.805223e-13
```

Fitted values:

```
Yhat=mod18$fitted.values
```

```
##           [,1]
## 1  -1.787459e-13
## 2  -1.941780e-13
## 3  -2.192690e-13
## 4  -2.028377e-13
## 5  -1.745271e-13
## 6  -1.820766e-13
## 7  -1.694200e-13
## 8  -1.563194e-13
## 9  -1.756373e-13
## 10 -1.851852e-13
## 11 -1.922906e-13
## 12 -2.049472e-13
## 13 -1.674216e-13
## 14 -1.780798e-13
## 15 -1.771916e-13
## 16 -1.638689e-13
## 17 -2.118306e-13
## 18 -1.718625e-13
## 19 -1.569855e-13
```

20 -1.911804e-13
21 -1.616485e-13
22 -1.638689e-13
23 -1.900702e-13
24 -1.669775e-13
25 -1.834088e-13
26 -1.625367e-13
27 -1.865175e-13
28 -2.027267e-13
29 -1.709743e-13
30 -1.763034e-13
31 -1.854072e-13
32 -2.065015e-13
33 -1.654232e-13
34 -1.714184e-13
35 -1.942890e-13
36 -1.700862e-13
37 -2.311484e-13
38 -1.814104e-13
39 -1.794120e-13
40 -1.680878e-13
41 -1.865175e-13
42 -1.869616e-13
43 -1.754152e-13
44 -1.900702e-13
45 -1.691980e-13
46 -1.540990e-13
47 -2.122746e-13
48 -1.807443e-13
49 -1.194600e-13
50 -1.749711e-13
51 -1.860734e-13
52 -1.536549e-13
53 -1.767475e-13
54 -1.803002e-13
55 -1.372236e-13
56 -1.740830e-13
57 -1.636469e-13
58 -1.443290e-13
59 -2.016165e-13
60 -1.594280e-13
61 -1.962874e-13
62 -1.758593e-13
63 -1.958433e-13
64 -1.763034e-13
65 -1.558753e-13
66 -1.665335e-13
67 -1.691980e-13
68 -1.749711e-13
69 -2.227107e-13
70 -2.002842e-13
71 -1.745271e-13
72 -1.860734e-13
73 -2.096101e-13

```
## 74 -1.776357e-13
## 75 -1.909584e-13
## 76 -1.696421e-13
## 77 -1.771916e-13
## 78 -1.754152e-13
## 79 -1.865175e-13
## 80 -1.576517e-13
## 81 -1.714184e-13
## 82 -1.523226e-13
## 83 -1.953993e-13
## 84 -1.838529e-13
## 85 -1.669775e-13
## 86 -1.629807e-13
## 87 -1.310063e-13
## 88 -1.785239e-13
## 89 -1.900702e-13
## 90 -1.922906e-13
## 91 -1.469935e-13
## 92 -1.403322e-13
## 93 -1.749711e-13
## 94 -1.225686e-13
## 95 -1.616485e-13
## 96 -1.563194e-13
## 97 -1.803002e-13
```

The three comparisons show that the manual matrix calculations are numerically equivalent to the results returned by `lm()`.

```
Yhat=predict(model8)
```

```
##           [,1]
## 1  -1.685319e-13
## 2  -1.979528e-13
## 3  -2.212674e-13
## 4  -2.018385e-13
## 5  -1.743050e-13
## 6  -1.822986e-13
## 7  -1.683098e-13
## 8  -1.532108e-13
## 9  -1.747491e-13
## 10 -1.842970e-13
## 11 -1.914024e-13
## 12 -2.041700e-13
## 13 -1.669775e-13
## 14 -1.774136e-13
## 15 -1.763034e-13
## 16 -1.627587e-13
## 17 -2.111644e-13
## 18 -1.705303e-13
## 19 -1.560974e-13
## 20 -1.898481e-13
## 21 -1.603162e-13
## 22 -1.620926e-13
```

23 -1.891820e-13
24 -1.652012e-13
25 -1.820766e-13
26 -1.612044e-13
27 -1.858513e-13
28 -2.022826e-13
29 -1.700862e-13
30 -1.740830e-13
31 -1.840750e-13
32 -2.047251e-13
33 -1.640910e-13
34 -1.703082e-13
35 -1.931788e-13
36 -1.691980e-13
37 -2.300382e-13
38 -1.809664e-13
39 -1.780798e-13
40 -1.671996e-13
41 -1.869616e-13
42 -1.856293e-13
43 -1.736389e-13
44 -1.891820e-13
45 -1.678657e-13
46 -1.527667e-13
47 -2.096101e-13
48 -1.794120e-13
49 -1.181277e-13
50 -1.731948e-13
51 -1.847411e-13
52 -1.518785e-13
53 -1.758593e-13
54 -1.785239e-13
55 -1.363354e-13
56 -1.727507e-13
57 -1.625367e-13
58 -1.429967e-13
59 -2.002842e-13
60 -1.585398e-13
61 -1.945111e-13
62 -1.745271e-13
63 -1.949552e-13
64 -1.749711e-13
65 -1.545430e-13
66 -1.656453e-13
67 -1.669775e-13
68 -1.736389e-13
69 -2.213785e-13
70 -1.989520e-13
71 -1.731948e-13
72 -1.851852e-13
73 -2.087219e-13
74 -1.763034e-13
75 -1.891820e-13
76 -1.674216e-13

```
## 77 -1.758593e-13
## 78 -1.740830e-13
## 79 -1.847411e-13
## 80 -1.558753e-13
## 81 -1.700862e-13
## 82 -1.501022e-13
## 83 -1.940670e-13
## 84 -1.816325e-13
## 85 -1.665335e-13
## 86 -1.625367e-13
## 87 -1.292300e-13
## 88 -1.776357e-13
## 89 -1.882938e-13
## 90 -1.914024e-13
## 91 -1.452172e-13
## 92 -1.385558e-13
## 93 -1.731948e-13
## 94 -1.207923e-13
## 95 -1.603162e-13
## 96 -1.536549e-13
## 97 -1.785239e-13
```

The fitted values obtained manually from the matrix calculation $X\hat{\beta}$ are also essentially identical to those returned by `predict(model8)`.

Confidence and Prediction Intervals

```
alpha = 0.05

beta.est = coef(model8)
stddev = sqrt(diag(vcov(model8)))

tcrit = qt(1 - alpha/2, df = model8$df.residual)

IC95 = cbind(
  estimation = beta.est,
  borne_inf = beta.est - tcrit * stddev,
  borne_sup = beta.est + tcrit * stddev
)

IC95
```

Confidence Intervals for Regression Coefficients

```
##           estimation    borne_inf    borne_sup
## (Intercept) 0.669399027 -1.906886346 3.245684401
## lcavol      0.587022881  0.412299613 0.761746149
## lweight     0.454460641  0.116597375 0.792323906
## age         -0.019637208 -0.041840687 0.002566271
## lbph        0.107054351 -0.009101413 0.223210115
## svi         0.766155885  0.280642108 1.251669662
```

```
## lcp          -0.105473570 -0.286343744 0.075396605
## gleason      0.045135964 -0.267791575 0.358063503
## pgg45        0.004525324 -0.004260852 0.013311499
```

```
confint(model8)
```

```
##              2.5 %      97.5 %
## (Intercept) -1.906886346 3.245684401
## lcavol       0.412299613 0.761746149
## lweight      0.116597375 0.792323906
## age         -0.041840687 0.002566271
## lbph        -0.009101413 0.223210115
## svi         0.280642108 1.251669662
## lcp         -0.286343744 0.075396605
## gleason     -0.267791575 0.358063503
## pgg45       -0.004260852 0.013311499
```

The manually computed 95% confidence intervals agree with those returned by `confint`. The interval for `age` contains 0, indicating that `age` is not statistically significant at the 5% level. By contrast, the interval for `lcavol` does not contain 0, indicating statistical significance. Applying the same reasoning to the remaining predictors identifies `lcavol`, `lweight`, and `svi` as significant variables.

```
new.obs = data.frame(
  lcavol = 0.75,
  lweight = 3.43,
  age = 62,
  lbph = -1.39,
  svi = 0,
  lcp = -1.39,
  gleason = 6,
  pgg45 = 0
)

predict(model8, newdata = new.obs, interval = "confidence", level = 0.95)
```

Confidence Interval for the Mean Response

```
##      fit      lwr      upr
## 1 1.719578 1.45585 1.983305
```

```
x0 = c(1, 0.75, 3.43, 62, -1.39, 0, -1.39, 6, 0)

yhat = sum(x0 * coef(model8))

se.mean = sqrt(t(x0) %*% vcov(model8) %*% x0)

tcrit = qt(0.975, df = model8$df.residual)
```

```
IC = c(
  fit = yhat,
  lwr = yhat - tcrit * se.mean,
  upr = yhat + tcrit * se.mean
)

IC
```

Manual Confidence-Interval Calculation

```
##      fit      lwr      upr
## 1.719578 1.455850 1.983305
```

```
predict(
  model8,
  newdata = data.frame(
    lcavol = 0.75,
    lweight = 3.43,
    age = 62,
    lbph = -1.39,
    svi = 0,
    lcp = -1.39,
    gleason = 6,
    pgg45 = 0
  ),
  interval = "prediction",
  level = 0.95
)
```

Prediction Interval for an Individual Observation

```
##      fit      lwr      upr
## 1 1.719578 0.2872602 3.151895
```

The prediction interval is wider than the confidence interval because it targets an individual future value of `lpsa` rather than the mean response at the specified covariate values. It therefore incorporates both parameter-estimation uncertainty and residual variability. Although the predicted mean is approximately 1.72, an individual value can plausibly range from 0.287 to 3.152 at the 95% level.

Hypothesis Testing and Variable Selection

Overall Significance of the Regression We use an F-test to determine whether the regression model is globally significant. The hypotheses are:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_8 = 0$$

against

$$H_1 : \text{at least one coefficient } \beta_j \text{ is non-zero.}$$

```
SCRH1= sum(model8$resid^2 )
SCRH0 = sum((Prostate$lpsa-mean(Prostate$lpsa))^2)

F= ((SCRH0-SCRH1)/(p-1))/(SCRH1/(n-p)); F
```

```
## [1] 20.86129
```

```
qf(1-alpha,p-1,model8$df.residual)
```

```
## [1] 2.045414
```

```
pf(F,p-1,model8$df.residual,lower.tail=FALSE)
```

```
## [1] 2.244848e-17
```

The p-value is far below 0.05, so the regression is globally significant.

```
model1=lm(lpsa~lcavol,data=Prostate)
anova(model1,model8)
```

Comparing a Single-Predictor Model with the Full Model

```
## Analysis of Variance Table
##
## Model 1: lpsa ~ lcavol
## Model 2: lpsa ~ lcavol + lweight + age + lbph + svi + lcp + gleason +
##      pgg45
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      95 58.915
## 2      88 44.163   7    14.752 4.1992 0.0004916 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
SCR_reduit = sum(model1$resid^2)
p_reduit = length(coef(model1))
q = p - p_reduit
Fobs = ((SCR_reduit - SCRH1) / q) /
      (SCRH1 / (n - p))
pvalue = 1 - pf(Fobs, df1 = q, df2 = n - p)

Fobs
```

Sequential ANOVA and Reduced-Model Selection

```
## [1] 4.199191
```

```
pvalue
```

```
## [1] 0.000491586
```

```
anova(model8)
```

```
## Analysis of Variance Table
##
## Response: lpsa
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## lcavol      1 69.003   69.003  137.4960 < 2.2e-16 ***
## lweight     1  5.948    5.948   11.8529 0.0008833 ***
## age         1  0.420    0.420    0.8369 0.3627907
## lbph        1  1.069    1.069    2.1302 0.1479844
## svi         1  5.952    5.952   11.8594 0.0008806 ***
## lcp         1  0.129    0.129    0.2576 0.6130570
## gleason     1  0.707    0.707    1.4097 0.2382955
## pgg45       1  0.526    0.526    1.0477 0.3088513
## Residuals  88 44.163    0.502
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The sequential analysis of variance for the full model indicates that `lcavol`, `lweight`, and `svi` contribute significantly to explaining `lpsa`. By contrast, `age`, `lbph`, `lcp`, `gleason`, and `pgg45` do not appear significant once the other variables are taken into account.

We therefore consider the reduced model:

$$lpsa \sim lcavol + lweight + svi.$$

```
lmf=lm(lpsa~lcavol+lweight+svi,data=Prostate)
anova(lmf,model8)
```

```
## Analysis of Variance Table
##
## Model 1: lpsa ~ lcavol + lweight + svi
## Model 2: lpsa ~ lcavol + lweight + age + lbph + svi + lcp + gleason +
##           pgg45
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      93 47.785
## 2      88 44.163   5    3.6218 1.4434 0.2167
```

We compare this reduced model with the full model. The test asks whether the removed variables—`age`, `lbph`, `lcp`, `gleason`, and `pgg45`—provide significant additional information beyond `lcavol`, `lweight`, and `svi`.

The hypotheses are:

$$H_0 : \beta_{age} = \beta_{lbph} = \beta_{lcp} = \beta_{gleason} = \beta_{pgg45} = 0$$

against:

H_1 : at least one of these coefficients is non-zero.

The comparison `anova(lmf, model8)` gives a p-value of approximately 0.217. Since this is greater than 0.05, we do not reject H_0 . The removed variables therefore do not significantly improve the model.

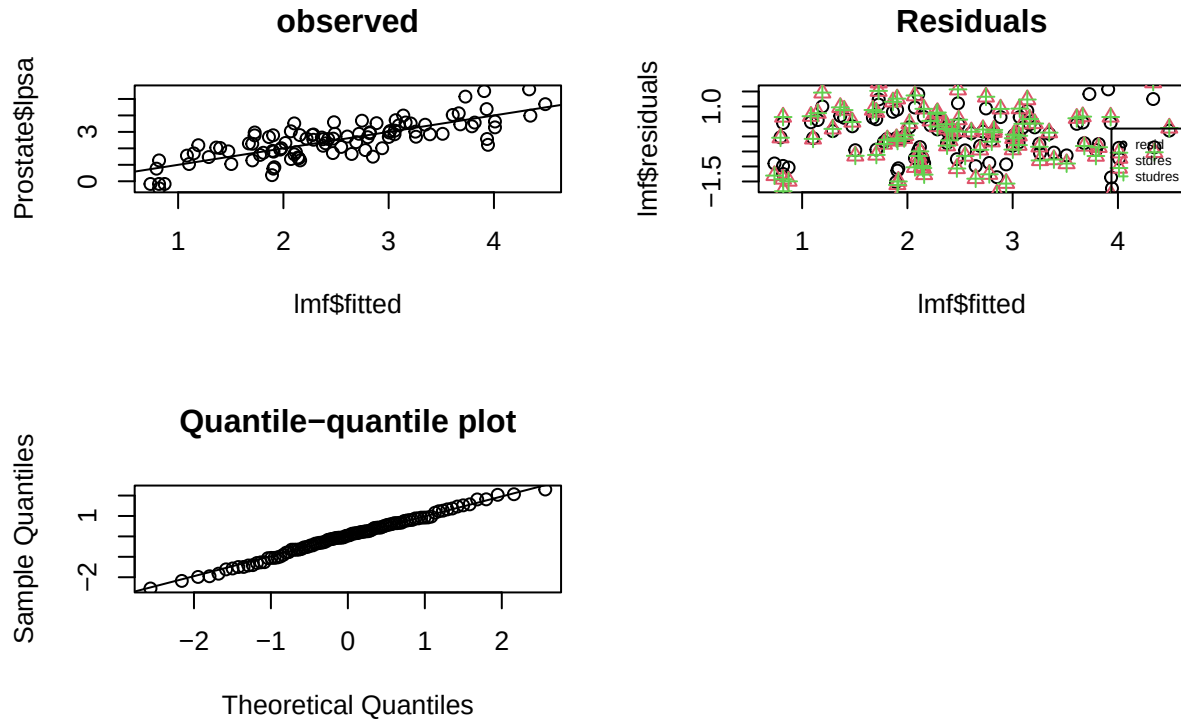
We retain the reduced model:

$$lpsa \sim lcavol + lweight + svi.$$

Model Diagnostics

```
#fenetre 2x2
par(mfrow=c(2,2), oma=c(0,0,2,0))
#graphe des observés
plot(lmf$fitted,Prostate$lpsa,main="observed")
abline(0,1)
#graphe des résidus
plot(lmf$fitted,lmf$residuals,main="Residuals")
library(MASS)
points(lmf$fitted,stdres(lmf), col=2,pch=2 )
points(lmf$fitted,studres(lmf),col=3,pch=3)
abline(h=2)
abline(h=-2)
legend("bottomright",c("resid", "stdres", "studres"),col=1:3, pch=1:3,
cex=0.5)
#graphe quantile
qqnorm(studres(lmf),main="Quantile-quantile plot")
qqline(stdres(lmf))
#titre
title(main="Validation en régression linéaire multiple", outer=TRUE)
```

Validation en régression linéaire multiple

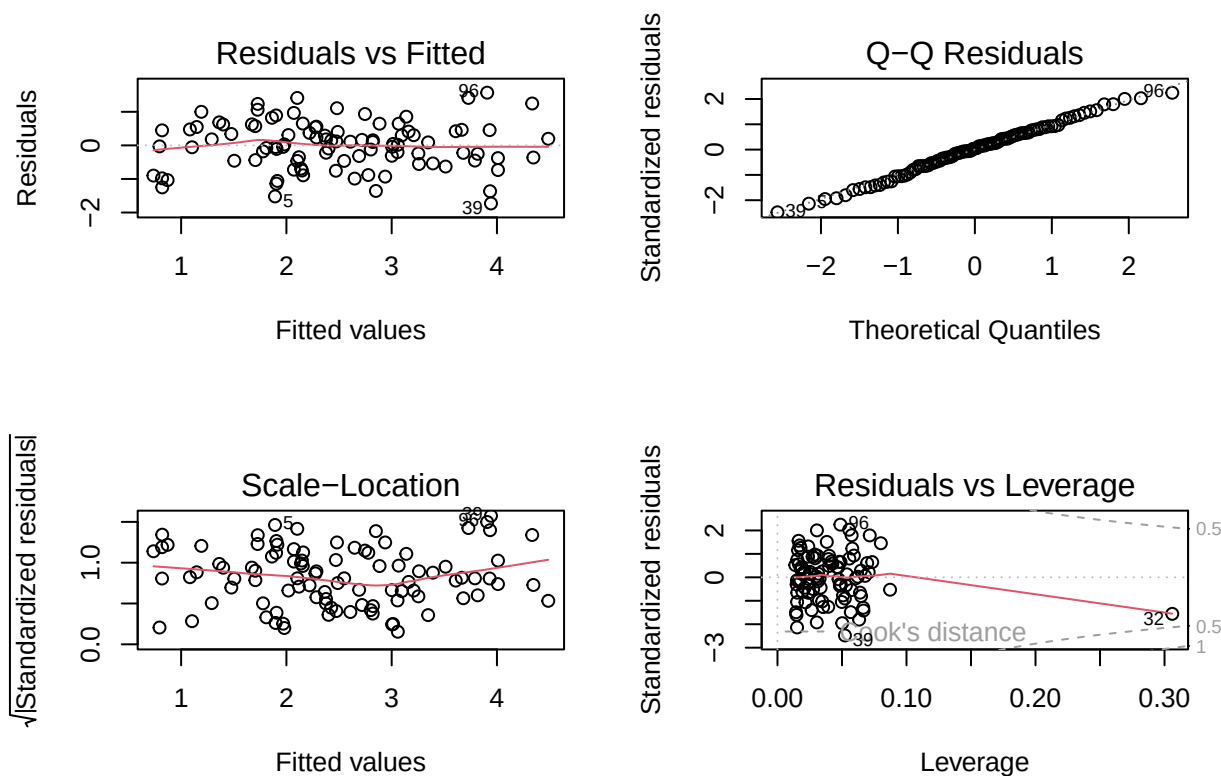


The first plot compares observed `lpsa` values with the fitted values from the model. The relationship is clearly increasing, indicating that the model captures a substantial part of the variation in `lpsa`. The points are not perfectly aligned with the identity line, however, so some variability remains unexplained.

The residuals-versus-fitted plot is used to assess whether residuals are randomly distributed around 0. Here, the residuals appear broadly scattered around 0 without a strong systematic pattern, suggesting that the linearity assumption is reasonable. A few observations are relatively distant from 0, but none appears extreme; most residuals are approximately between -2 and 2.

The Q-Q plot is used to assess residual normality. The points are generally close to the reference line, suggesting that the residual distribution is approximately normal.

```
par(mfrow=c(2,2), oma=rep(0,4))  
plot(lmf)
```



The residuals are generally centered around 0. The red smooth curve is nearly horizontal, with only mild curvature, suggesting that the linearity assumption is acceptable even though some variation remains unexplained.

In the Q-Q plot, most points lie close to the reference line. This indicates that the standardized residuals are approximately normally distributed. There are small deviations in the tails, particularly for observations 39, 5, and 96, but they do not appear severe.

The Scale-Location plot assesses the constant-variance assumption. The points are fairly dispersed, and the red curve is not perfectly horizontal: it decreases and then rises slightly. This may indicate mild heteroscedasticity, but the effect does not appear substantial.

Most observations have low leverage. Observation 32 has higher leverage than the others and may therefore exert more influence on the fitted model. However, it does not clearly cross the Cook's-distance contours, so it is not necessarily problematic.

```
model8 = lm(lpsa ~ ., data = Prostate)

model_aic = step(model8, direction = "both", trace = 0)
model_bic = step(model8, direction = "both", k = log(nrow(Prostate)), trace = 0)

model_aic

##
## Call:
## lm(formula = lpsa ~ lcaVOL + lweight + age + lbph + svi, data = Prostate)
##
```

```
## Coefficients:
## (Intercept)      lcavol      lweight      age      lbph      svi
##      0.95102      0.56561      0.42369     -0.01489      0.11184      0.72096
```

```
model_bic
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol + lweight + svi, data = Prostate)
##
## Coefficients:
## (Intercept)      lcavol      lweight      svi
##      -0.2681      0.5516      0.5085      0.6662
```

AIC selects the model $\text{lpsa} \sim \text{lcavol} + \text{lweight} + \text{age} + \text{lbph} + \text{svi}$, while BIC selects the more parsimonious model $\text{lpsa} \sim \text{lcavol} + \text{lweight} + \text{svi}$.

Case Study 2 — ANCOVA for Infusion-System Alarm Delays

Data and Experimental Design

```
perf=read.table("perfusion.txt",header=TRUE, row.names=1,skip=1)
perf$systeme = as.factor(perf$systeme)
summary(perf)
```

```
##      delai      debit      systeme
## Min.   : 185.0   Min.    : 2.500   S1:24
## 1st Qu.: 345.0   1st Qu.: 6.875   S2:24
## Median : 582.5   Median :11.250   S3:24
## Mean   : 838.9   Mean    :12.188   S4:24
## 3rd Qu.: 981.2   3rd Qu.:16.250
## Max.   :3460.0   Max.    :25.000
```

```
head(perf)
```

```
##      delai debit systeme
## 1  1970   2.5      S1
## 2  2775   2.5      S2
## 3  3460   2.5      S3
## 4  1965   2.5      S4
## 5  1990   2.5      S1
## 6  2790   2.5      S2
```

```
dim(perf)
```

```
## [1] 96  3
```

```
str(perf)
```

```
## 'data.frame': 96 obs. of 3 variables:
## $ delai : int 1970 2775 3460 1965 1990 2790 3440 1985 1995 2785 ...
## $ debit : num 2.5 2.5 2.5 2.5 2.5 2.5 2.5 2.5 2.5 2.5 ...
## $ systeme: Factor w/ 4 levels "S1","S2","S3",...: 1 2 3 4 1 2 3 4 1 2 ...
```

Exploratory Analysis

```
table(perf$systeme, perf$debit)
```

Experimental Balance

```
##
##      2.5 5 7.5 10 12.5 15 20 25
## S1   3 3 3 3 3 3 3 3
## S2   3 3 3 3 3 3 3 3
## S3   3 3 3 3 3 3 3 3
## S4   3 3 3 3 3 3 3 3
```

```
tapply(perf$delai, list(perf$systeme, perf$debit), mean)
```

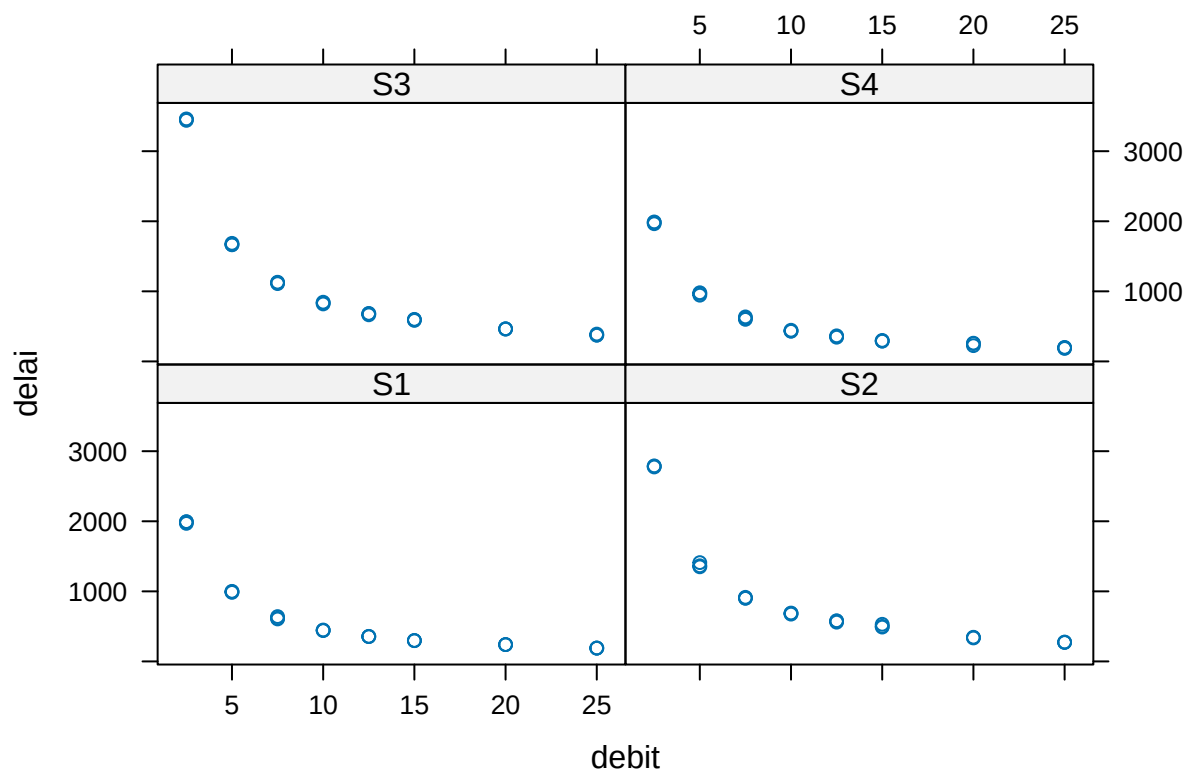
```
##      2.5      5      7.5      10      12.5      15      20      25
## S1 1985.000 993.3333 620.3333 443.6667 355.6667 297.3333 240.0000 190.0000
## S2 2783.333 1375.0000 906.6667 683.3333 571.6667 506.6667 340.0000 273.3333
## S3 3451.667 1671.6667 1121.6667 833.3333 676.6667 593.3333 463.3333 380.0000
## S4 1980.000 965.0000 616.6667 436.6667 355.0000 293.3333 246.6667 193.3333
```

There are three measurements for each system-flow-rate combination. The experimental design is therefore balanced: every system is observed at the same flow rates and with the same number of replicates.

The table of means shows that, for every system, alarm delay decreases as flow rate increases. This is consistent with the expected behavior: higher flow rates lead to faster alarm activation. The systems also behave differently. System *S3* has the largest mean delays, system *S2* has intermediate delays, and systems *S1* and *S4* have very similar mean delays.

```
library(lattice)
xyplot(delai~debit|systeme,data=perf)
```

Visualization by System



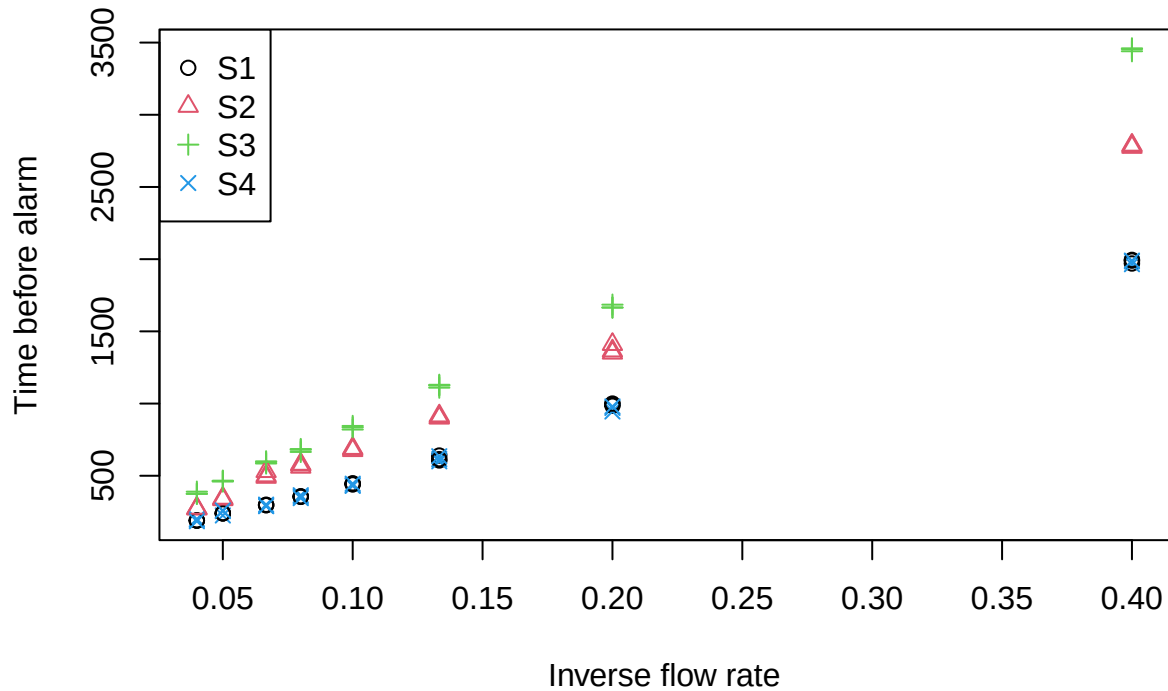
```
par(mfrow=c(3,1))
```

Linearizing the Relationship The flow rate appears to vary approximately as the inverse of the delay. Using $1/\text{debit}$ as a transformed predictor therefore makes it possible to study the relationship using a linear model.

```
perf$invdebit=1/perf$debit
par(mfrow=c(1,1))
cols_systeme = as.numeric(perf$systeme)
plot(perf$invdebit,perf$delai,col=cols_systeme,
     pch=cols_systeme,
     main="Alarm time for different infusion systems",
     xlab="Inverse flow rate",ylab="Time before alarm")
legend("topleft",c("S1", "S2", "S3", "S4"),pch=c(1:4),col=c(1:4))
```

Visualization After Transformation

Alarm time for different infusion systems



ANCOVA Model

We use the following ANCOVA model:

$$\text{delai}_{ik} = (\mu + a_k) + (\beta + b_k) \text{invdebit}_i + \varepsilon_{ik}$$

where $\text{invdebit}_i = \frac{1}{\text{debit}_i}$, μ is the average intercept, β is the average slope associated with inverse flow rate, a_k is the effect of system k on the intercept, b_k is the effect of system k on the slope, and ε_{ik} is the error term.

Model Estimation

```
Mg=lm(delai~systeme*invdebit,data=perf)
summary(Mg)
```

```
##
## Call:
## lm(formula = delai ~ systeme * invdebit, data = perf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.662 -19.161  -3.496   21.495   64.621
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -33.53225     7.88743  -4.251 5.27e-05 ***
## systemeS2      37.17563    11.15451   3.333 0.001259 **
## systemeS3      40.41060    11.15451   3.623 0.000487 ***
## systemeS4     -0.08376    11.15451  -0.008 0.994026
## invdebit      5040.73955    45.28203 111.319 < 2e-16 ***
## systemeS2:invdebit 1885.29121    64.03846  29.440 < 2e-16 ***
## systemeS3:invdebit 3498.17622    64.03846  54.626 < 2e-16 ***
## systemeS4:invdebit -35.51081    64.03846  -0.555 0.580627
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 24.75 on 88 degrees of freedom
## Multiple R-squared:  0.999, Adjusted R-squared:  0.999
## F-statistic: 1.297e+04 on 7 and 88 DF, p-value: < 2.2e-16
```

```
anova(Mg)
```

```
## Analysis of Variance Table
##
## Response: delai
##              Df    Sum Sq Mean Sq F value    Pr(>F)
## systeme        3  4439230  1479743    2415 < 2.2e-16 ***
## invdebit        1  48620012  48620012   79349 < 2.2e-16 ***
## systeme:invdebit  3   2582679    860893    1405 < 2.2e-16 ***
## Residuals      88     53921      613
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In the theoretical ANCOVA formulation, the identifiability conditions can be expressed using sum-to-zero constraints:

$$\sum_{k=1}^4 a_k = 0 \quad \text{and} \quad \sum_{k=1}^4 b_k = 0.$$

With R's `lm()` function, one factor level is instead used as the reference category, here S_1 . The coefficients displayed by R therefore represent differences relative to S_1 .

```
Mg = lm(delai ~ systeme * invdebit, data = perf)
```

```
plot(
  perf$invdebit, perf$delai,
  col = cols_systeme,
  pch = cols_systeme,
  xlab = "Inverse du débit",
  ylab = "Délai",
  main = "Droites d'ajustement selon le système"
)
```

```
co = coef(Mg)
```

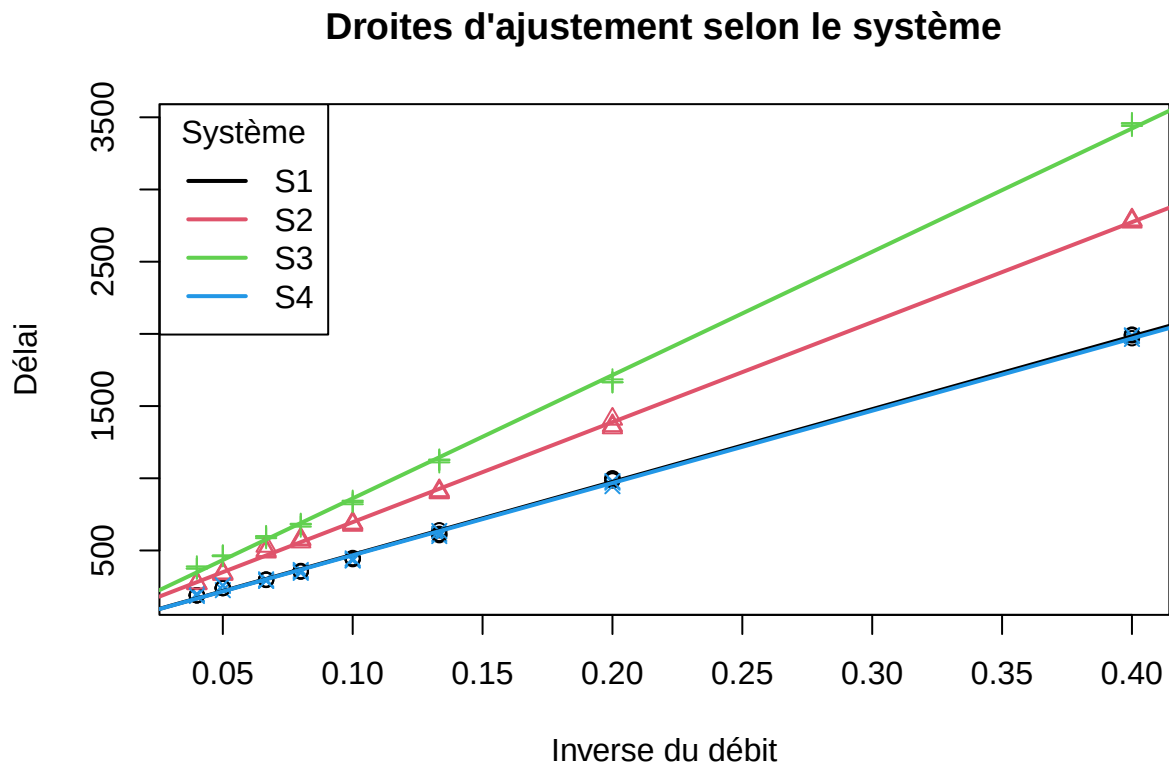
```

abline(co[1], co[5], col = 1, lwd = 2)

for (i in 2:4) {
  abline(co[1] + co[i], co[5] + co[4 + i], col = i, lwd = 2)
}

legend(
  "topleft",
  legend = c("S1", "S2", "S3", "S4"),
  col = 1:4,
  lwd = 2,
  title = "Système"
)

```



```

Mgbis=lm(delai~systeme+systeme:invdebit,data=perf)
summary(Mgbis)

```

```

##
## Call:
## lm(formula = delai ~ systeme + systeme:invdebit, data = perf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.662 -19.161  -3.496   21.495   64.621

```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -33.53225     7.88743  -4.251 5.27e-05 ***
## systemeS2      37.17563    11.15451   3.333 0.001259 **
## systemeS3      40.41060    11.15451   3.623 0.000487 ***
## systemeS4      -0.08376    11.15451  -0.008 0.994026
## systemeS1:invdebit 5040.73955  45.28203 111.319 < 2e-16 ***
## systemeS2:invdebit 6926.03076  45.28203 152.953 < 2e-16 ***
## systemeS3:invdebit 8538.91577  45.28203 188.572 < 2e-16 ***
## systemeS4:invdebit 5005.22874  45.28203 110.535 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 24.75 on 88 degrees of freedom
## Multiple R-squared:  0.999, Adjusted R-squared:  0.999
## F-statistic: 1.297e+04 on 7 and 88 DF, p-value: < 2.2e-16
```

```
anova(Mgbis)
```

```
## Analysis of Variance Table
##
## Response: delai
##              Df    Sum Sq Mean Sq F value    Pr(>F)
## systeme        3  4439230  1479743    2415 < 2.2e-16 ***
## systeme:invdebit 4 51202691 12800673   20891 < 2.2e-16 ***
## Residuals      88    53921     613
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The two model parameterizations are equivalent in terms of fitted values. The difference is that the second formulation does not include a separate overall `invdebit` effect: the slopes are represented directly through the system-specific terms.

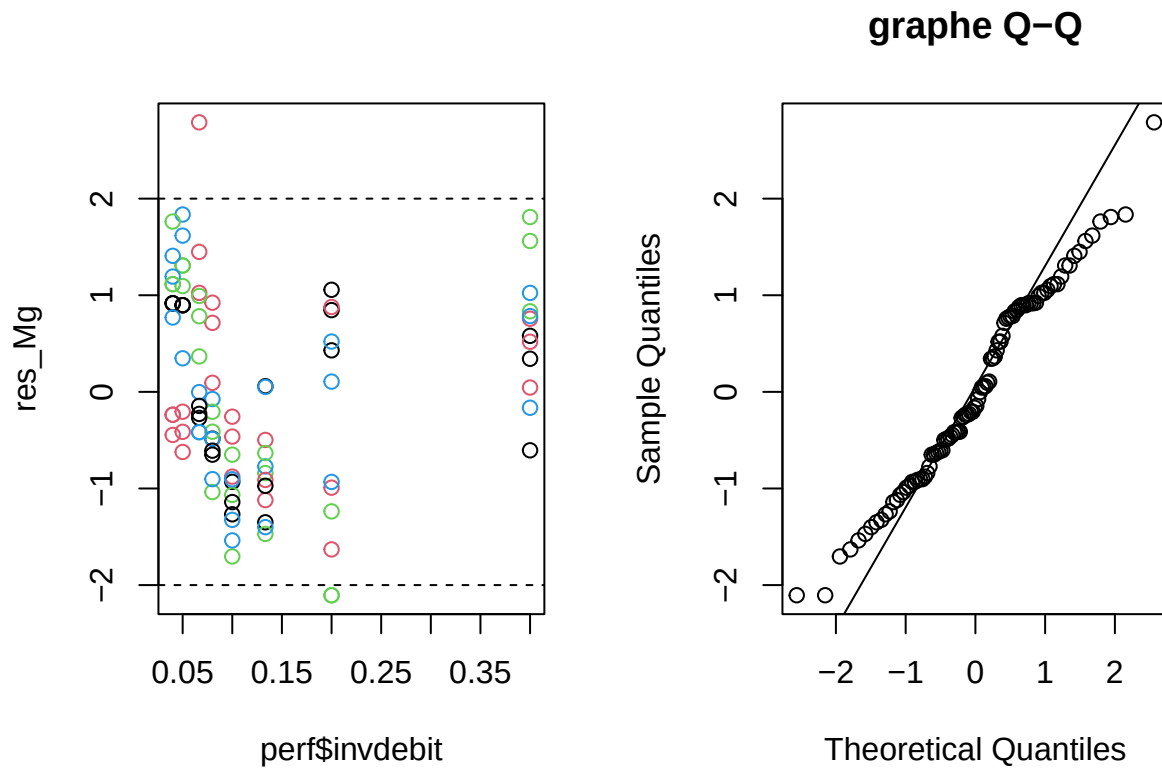
Model Validation

```
par(mfrow=c(1,2),oma=c(0,0,0,0))

res_Mg = rstudent(Mg)

plot(perf$invdebit,res_Mg,
     col=cols_systeme)
abline(h=2,lty=2); abline(h=-2,lty=2)

qqnorm(res_Mg,main="graphe Q-Q")
qqline(res_Mg)
```



```
shapiro.test(res_Mg)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res_Mg
## W = 0.97899, p-value = 0.1263
```

The Shapiro-Wilk test gives $W = 0.9789$ and a p-value of $0.1263 > 0.05$. We therefore do not reject the null hypothesis of normality for the studentized residuals of the full model Mg .

Nested Model Comparison

We first test whether all systems can be represented by the same regression line.

The null hypothesis corresponds to a common line for all systems:

$$H_0 : a_1 = a_2 = a_3 = a_4 = 0 \quad \text{and} \quad b_1 = b_2 = b_3 = b_4 = 0.$$

The alternative hypothesis is:

$$H_1 : \text{at least one parameter } a_k \text{ or } b_k \text{ is non-zero.}$$

The test statistic is:

$$F = \frac{(SCR_0 - SCR_g)/(p_g - p_0)}{SCR_g/(n - p_g)}.$$

Under H_0 ,

$$F \sim \mathcal{F}(p_g - p_0, n - p_g).$$

```
M0 = lm(delai ~ invdebit, data = perf)

SCR0 = sum(M0$resid^2)
SCRg = sum(Mg$resid^2)

n = nrow(perf)
p0 = length(coef(M0))
pg = length(coef(Mg))

Fobs = ((SCR0 - SCRg) / (pg - p0)) / (SCRg / (n - pg))

pvalue = pf(Fobs, df1 = pg - p0, df2 = n - pg, lower.tail = FALSE)

Fobs
```

```
## [1] 1909.981
```

```
pvalue
```

```
## [1] 6.539853e-91
```

```
anova(M0, Mg)
```

```
## Analysis of Variance Table
##
## Model 1: delai ~ invdebit
## Model 2: delai ~ systeme * invdebit
##   Res.Df    RSS Df Sum of Sq   F    Pr(>F)
## 1      94 7075830
## 2      88  53921  6   7021909 1910 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The resulting p-value is far below 0.05 (6.5×10^{-91}). We therefore reject the hypothesis that all systems share the same regression line. The infusion systems do not follow the same relationship between alarm delay and inverse flow rate, so the full ANCOVA model is preferred to the reduced model with a single common line.

We next compare several constrained submodels with the full model.

```
M2 = lm(delai ~ systeme:invdebit, data = perf); summary(M2)
```

```
##
## Call:
## lm(formula = delai ~ systeme:invdebit, data = perf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -47.788 -22.259   1.763  17.557  77.190
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -14.157      4.382   -3.231  0.00172 **
## systemeS1:invdebit 4955.325    37.573  131.884 < 2e-16 ***
## systemeS2:invdebit 7004.499    37.573  186.421 < 2e-16 ***
## systemeS3:invdebit 8631.645    37.573  229.727 < 2e-16 ***
## systemeS4:invdebit 4919.445    37.573  130.929 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.5 on 91 degrees of freedom
## Multiple R-squared:  0.9988, Adjusted R-squared:  0.9987
## F-statistic: 1.839e+04 on 4 and 91 DF,  p-value: < 2.2e-16
```

```
anova(M2, Mg)
```

```
## Analysis of Variance Table
##
## Model 1: delai ~ systeme:invdebit
## Model 2: delai ~ systeme * invdebit
##   Res.Df  RSS Df Sum of Sq    F    Pr(>F)
## 1      91 68827
## 2      88 53921  3    14906 8.1088 7.91e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
M3 = lm(delai ~ systeme + invdebit, data = perf);summary(M3)
```

```
##
## Call:
## lm(formula = delai ~ systeme + invdebit, data = perf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -368.90  -78.23  -12.18   83.42  612.97
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -212.355     40.507   -5.242 1.02e-06 ***
## systemeS2     289.333     49.137    5.888 6.46e-08 ***
## systemeS3     508.292     49.137   10.344 < 2e-16 ***
## systemeS4     -4.833     49.137   -0.098  0.922
## invdebit     6377.729    155.690   40.964 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 170.2 on 91 degrees of freedom
## Multiple R-squared:  0.9527, Adjusted R-squared:  0.9506
## F-statistic: 457.8 on 4 and 91 DF,  p-value: < 2.2e-16
```

```
anova(M3, Mg)
```

```
## Analysis of Variance Table
##
## Model 1: delai ~ systeme + invdebit
## Model 2: delai ~ systeme * invdebit
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      91 2636600
## 2      88  53921  3   2582679 1405 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
perf$systeme14 = perf$systeme
perf$systeme14[perf$systeme == "S4"] = "S1"
M4 = lm(delai ~ systeme14 * invdebit, data = perf);summary(M4)
```

```
##
## Call:
## lm(formula = delai ~ systeme14 * invdebit, data = perf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.662 -20.481  -3.778   22.312   64.621
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -33.574      5.539  -6.062 3.10e-08 ***
## systeme14S2      37.218      9.594   3.879 0.000199 ***
## systeme14S3      40.452      9.594   4.217 5.89e-05 ***
## invdebit      5022.984     31.799 157.962 < 2e-16 ***
## systeme14S2:invdebit 1903.047     55.077  34.552 < 2e-16 ***
## systeme14S3:invdebit 3515.932     55.077  63.837 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 24.58 on 90 degrees of freedom
## Multiple R-squared:  0.999, Adjusted R-squared:  0.999
## F-statistic: 1.841e+04 on 5 and 90 DF,  p-value: < 2.2e-16
```

```
anova(M4, Mg)
```

```
## Analysis of Variance Table
##
## Model 1: delai ~ systeme14 * invdebit
## Model 2: delai ~ systeme * invdebit
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      90 54390
## 2      88 53921  2    468.75 0.3825 0.6833
```

```

perf$ord23 = perf$systeme
perf$ord23[perf$systeme == "S2"] = "S3"
M5 = lm(delai ~ ord23 + invdebit + systeme14:invdebit, data = perf)
summary(M5)

##
## Call:
## lm(formula = delai ~ ord23 + invdebit + systeme14:invdebit, data = perf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.470 -18.989  -4.326   20.024   63.479
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -31.157      6.564  -4.747 7.75e-06 ***
## ord23S3         36.418      8.581   4.244 5.33e-05 ***
## ord23S4        -4.833      7.082  -0.683  0.497
## invdebit       5022.984     31.732 158.295 < 2e-16 ***
## invdebit:systeme14S2 1895.916     49.265  38.484 < 2e-16 ***
## invdebit:systeme14S3 3523.062     49.265  71.512 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 24.53 on 90 degrees of freedom
## Multiple R-squared:  0.999, Adjusted R-squared:  0.999
## F-statistic: 1.849e+04 on 5 and 90 DF, p-value: < 2.2e-16

anova(M5, Mg)

```

```

## Analysis of Variance Table
##
## Model 1: delai ~ ord23 + invdebit + systeme14:invdebit
## Model 2: delai ~ systeme * invdebit
##   Res.Df  RSS Df Sum of Sq    F Pr(>F)
## 1      90 54161
## 2      88 53921  2    239.95 0.1958 0.8225

```

We compare each submodel with the full model:

$$Mg : \text{delai} \sim \text{systeme} * \text{invdebit}.$$

The full model allows each system to have its own regression line.

For model $M2$, we assume a common intercept across systems but allow different slopes:

$$M2 : \text{delai} \sim \text{systeme} : \text{invdebit}$$

Compared with the full model Mg , the test gives a p-value of approximately 7.91×10^{-5} . Since this is below 0.05, we reject $M2$. The intercepts cannot be treated as identical across all systems.

For model $M3$, we assume equal slopes across systems:

$M3 : \text{delai} \sim \text{systeme} + \text{invdebit}.$

The comparison with Mg gives a p-value below 2.2×10^{-16} . We therefore strongly reject $M3$: the slopes differ across systems.

For model $M4$, we assume that systems $S1$ and $S4$ share the same regression line, while the others remain distinct. The comparison with Mg gives a p-value of approximately 0.683, which is above 0.05. We therefore do not reject $M4$, and the data are consistent with $S1$ and $S4$ sharing the same regression line.

For model $M5$, we retain the assumption $S1 = S4$ and additionally impose equal intercepts for $S2$ and $S3$. The comparison with Mg gives a p-value of approximately 0.82, again well above 0.05. We therefore do not reject $M5$.

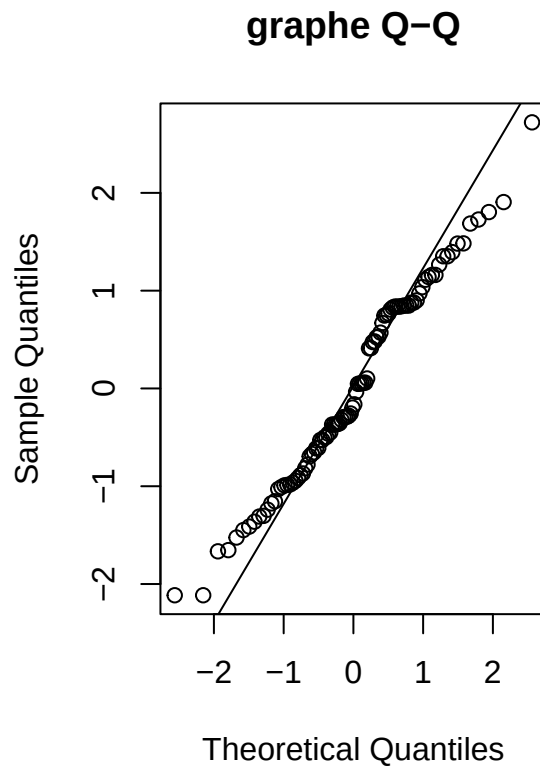
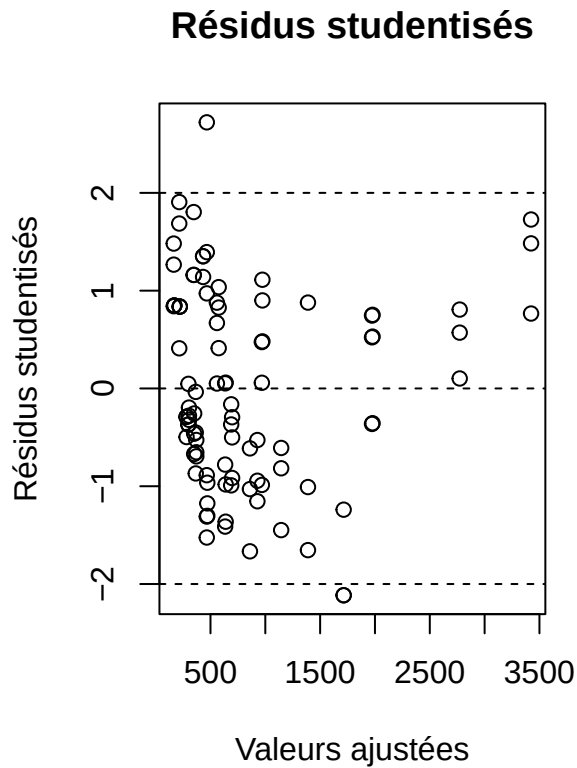
The selected model is therefore $M5$.

```
res_M5 = rstudent(M5)

par(mfrow=c(1,2),oma=c(0,0,0,0))

plot(fitted(M5), res_M5,
     xlab = "Valeurs ajustées",
     ylab = "Résidus studentisés",
     main = "Résidus studentisés")
abline(h=0,lty=2)
abline(h=2,lty=2)
abline(h=-2,lty=2)

qqnorm(res_M5,main="graphe Q-Q")
qqline(res_M5)
```



```
shapiro.test(res_M5)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res_M5
## W = 0.97897, p-value = 0.1258
```

The Shapiro-Wilk test applied to the studentized residuals of $M5$ gives a p-value of approximately 0.126, which is above 0.05. We therefore do not reject the normality assumption. The diagnostic plots do not reveal any major issue; apart from the tails, the Q-Q plot follows the reference line reasonably well.

We therefore retain $M5$ as the final model.

Confidence Interval for the Difference Between Systems

For a flow rate of 5 ml/h, $\text{invdebit} = 1/5 = 0.2$. We compute a 95% confidence interval for the difference of interest:

```
L = rep(0, length(coef(M5)))
names(L) = names(coef(M5))

L["invdebit:système14S2"] = 0.2
L["invdebit:système14S3"] = -0.2
```

```

diff.est = sum(L * coef(M5))

se.diff = sqrt(t(L) %*% vcov(M5) %*% L)

tcrit = qt(0.975, df = M5$df.residual)

IC.diff = c(
  estimation = diff.est,
  borne_inf = diff.est - tcrit * se.diff,
  borne_sup = diff.est + tcrit * se.diff
)

IC.diff

```

```

## estimation borne_inf borne_sup
## -325.4292 -341.5831 -309.2752

```

The 95% confidence interval is:

$$IC_{95\%} = [-341.58 ; -309.28].$$

The interval is entirely negative and does not contain 0. We therefore conclude that, at a flow rate of 5 ml/h, the mean alarm delay for system *S2* is significantly lower than for system *S3*. More precisely, system *S2* triggers the alarm approximately 325 time units earlier than system *S3*, on average.

Conclusion

Across the two case studies, the analysis combines model formulation, estimation, statistical inference, model selection, and diagnostic validation. The prostate-cancer analysis supports a reduced linear model based on *lcavol*, *lweight*, and *svi*, while the infusion-system analysis supports the constrained ANCOVA model *M5* after comparison with the full model and several nested alternatives.